2.3 Focus of a Parabola

Algebra II 2.3 · Student sheet

Name _____ Period ____ Date _____

I can…

- identify the focus, directrix, and axis of symmetry of a parabola.
- write equations of parabolas from the focus, directrix, or vertex.
- solve real-life problems involving the focus of a parabola.

Vocabulary · fill in the missing word

Parabola — the set of all points _____ distant from a point (the focus) and a line (the directrix)

Focus — the fixed point inside the _____

Directrix — the fixed line outside the _____

Vertical axis (vertex at origin) — $y = (\frac{1}{4p})x^2$: focus (0, p), _____ $y = -p$

Horizontal axis (vertex at origin) — $x = (\frac{1}{4p})y^2$: focus (p, 0), _____ $x = -p$

Part 1 · Equal distances

Focus F(0, 2) and directrix $y = -2$. Check each point.

Point	Distance to F	Distance to $y = -2$	Equal?
(0, 0)			
(4, 2)			
(−4, 2)			
(2, 0.5)			
(8, 8)			
(2, 2)			

Plot the points that work and sketch the curve. Equation: _____

Part 2 · Notes: fill in the blanks

Every point on a parabola is the same distance from the _____ and the _____.

$y = (1/(4p))x^2$: focus _____, directrix _____.

$x = (1/(4p))y^2$: focus _____, directrix _____.

Part 3 · Examples (work them with the class)

Example 1

Identify the focus, directrix, and axis of symmetry.

a. $y = -(\frac{1}{12})x^2$ b. $3x^2 - 24y = 0$

Example 2

Identify the focus, directrix, and axis of symmetry of $x = \frac{1}{4}y^2$. Which way does it open?

Example 3

Write an equation of the parabola with vertex at the origin and

a. directrix $y = -5$. b. focus $(-2, 0)$.

Example 4

Write an equation of the parabola with vertex (3, 1) and focus (3, 4). Give the directrix.

Example 5

A satellite dish has a cross section modeled by $y = (\frac{1}{32})x^2$, with x and y in inches. How far from the vertex should the receiver be placed (at the focus)?

Part 4 · Partner practice

Partner 1

Identify the focus and directrix of $y = (\frac{1}{16})x^2$.

Partner 2

Write an equation of the parabola with vertex (0, 0) and focus (0, −1).